

BIOCHEMISTRY

1. BRAIN METABOLISM

➤ **Fuels used by brain:**

1-Glucose.

2-Ketone bodies.

3-Amino acid

- Nervous system closely depends on the availability of **glucose**, the lack of which quickly results in death.
- **Glucose** is brought to the brain by blood flowing through a special network of arteries and is quickly catabolized by the **glycolytic** and **tricarboxylic acid cycle** pathways to synthesize **energy**.
- It is also used in the synthesis of amino acids, nucleotides and NADPH (HMP pathway).
- **Glucose** can be converted into **glycogen** in the brain.
- The nerve tissue is capable of synthesizing glucose-6-phosphate in the **gluconeogenic pathway**
- Brain carbohydrate metabolism is thus in many ways similar to liver carbohydrate metabolism.
- **The astrocyte** constitutes the main cell implicated in this metabolism.

➤ **Brain Energy Metabolism**

- **Brain** is one of the most metabolically active organs in the body.

- The **brain** stores little **energy** as glycogen and relies almost entirely on circulating glucose for fuel.
- Inside neurons, glucose is metabolized by cytoplasm & mitochondria in a number of steps (**glycolysis & Krebs's cycle**) to produce cellular **energy**, as **ATP**.

➤ **Role of Oxygen to Brain**

- Having enough **oxygen** in the **brain** is crucial to brain function, and therefore, to the body as a whole.
- The **brain** requires a constant supply of both **oxygen** and **glucose**.
- Some brain cells start dying less than 5 minutes after their oxygen supply disappears.
- Without enough **oxygen** or with **low blood-oxygen** levels, lead to **brain** death.

➤ **Importance of Glucose**

1. Major source of energy for RBC and CNS.
2. The only source of energy for skeletal muscle during exercise.
3. Lipogenesis formation.
4. Krebs's cycle.

➤ **Fuel of brain**

A-General: use glucose

B-During Starvation: utilize ketone bodies

➤ **Amino Acid Content In Cerebral Tissue**

- The three **amino acids** with the highest levels in brain tissues are **GLUTAMIC ACID, GLUTAMINE, and ASPARTIC ACID.**
- Metabolism of glutamic acid and related substances plays a role in regulating brain functions

➤ **Fed Starvation Cycle**

Fate of glucose in Fed State

Fate of glucose in liver:

A- Make **Glycolysis** then **Kreb's Cycle**

B-**Excess glucose**: stored in liver as **Glycogen** which used to maintain blood glucose between meal

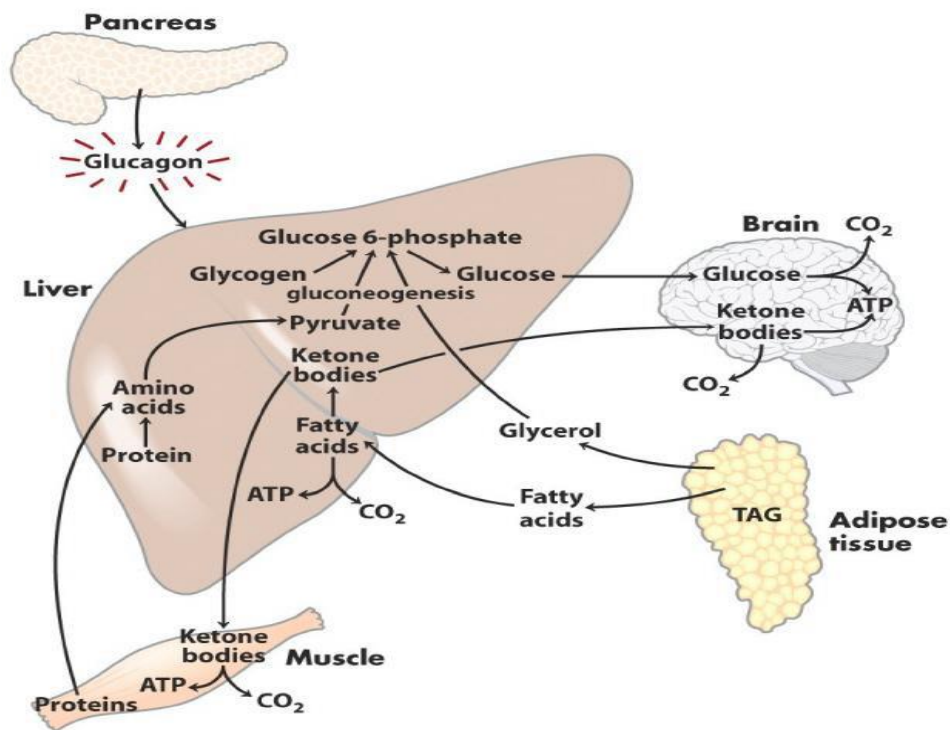
C-More **excess glucose**: converted to **TriAcylGlycerol (TAG)** synthesis

Fate of glucose in other tissues

- **Brain**: make **Glycolysis** then **Kreb's Cycle**
- **RBC**: make **Anaerobic Glycolysis** end by lactic acid
- **Muscle**:

1- Make **Glycolysis** then **Kreb's Cycle**

2- Excess glucose: stored in muscle as **Glycogen**



Fasting (Starvation) State

Energy produced by different food stuff

- **Carbohydrate:** 4 Kcal/g
- **Protein:** 4 Kcal/g
- **Fat:** 9 Kcal/g

➤ Brain in Starvation

- During first few day of starvation brain use glucose exclusively that provided from rapid breakdown of **muscle protein**
- After 3 week of starvation **ketone bodies** markedly increase and used as fuel for brain, this decreases breakdown of muscle protein